

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Microsoft IAD11, VA -- station KIAD, America/New_York
Committed pair OSM 1301654223 -> 897226575
 OSM way 1301654223 / Amazon IAD124
Facade gap 77.4 m between the two halls
Weather record 43,763 real hourly records from KIAD
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise
 0.3579 C at 140 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2025-03-11 (crossing)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 1 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 2 of 24 hours with 1 mode change. The reactive incumbent took 13 hours -- 11 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 12 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 2 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 13 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s)
 the incumbent broke its own switch budget 1 time(s) to stay safe
 12 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	4.262	18.0	4.077	1.041
01	mechanical	yes	switch budget	4.262	18.0	4.077	0.982
02	mechanical	yes	switch budget	3.982	18.0	2.967	1.151
03	mechanical	yes	switch budget	2.860	18.0	1.110	0.960
04	mechanical	yes	switch budget	2.877	18.0	0.747	0.980
05	mechanical	yes	switch budget	2.292	18.0	0.187	1.156
06	mechanical	yes	switch budget	0.932	18.0	-0.923	0.971
07	mechanical	yes	switch budget	2.597	18.0	0.187	1.625
08	mechanical	yes	switch budget	6.400	18.0	5.659	1.941
09	mechanical	yes	switch budget	9.939	18.0	11.418	2.346

10	mechanical	yes	switch budget	11.525	18.0	13.890	2.201
11	mechanical	yes	switch budget	15.456	18.0	17.785	1.780
12	mechanical	no	dry-bulb	18.197	18.0	19.441	1.315
13	mechanical	no	dry-bulb	20.099	18.0	21.110	1.172
14	mechanical	no	dry-bulb	22.203	18.0	22.017	1.281
15	mechanical	no	dry-bulb	22.904	18.0	22.780	1.127
16	mechanical	no	dry-bulb	22.916	18.0	22.780	1.089
17	mechanical	no	dry-bulb	22.662	18.0	22.220	1.385
18	mechanical	no	dry-bulb	22.081	18.0	21.110	1.272
19	mechanical	no	dry-bulb	20.726	18.0	18.891	1.390
20	mechanical	no	dry-bulb	19.811	18.0	17.785	1.453
21	mechanical	no	dry-bulb	18.441	18.0	15.561	1.633
22	FREE	yes	-	15.701	18.0	12.220	1.335
23	FREE	yes	-	15.726	18.0	13.341	1.120

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.262 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.262 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.982 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.860 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.877 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.292 C

against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.932 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.597 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.400 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.939 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.525 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.456 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.197 C against a 18.0 C limit, so it fails by 0.197 C. It would take a limit 0.197 C higher, or a bound 0.197 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.099 C against a 18.0 C limit, so it fails by 2.099 C. It would take a limit 2.099 C higher, or a bound 2.099 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.203 C against a 18.0 C limit, so it

fails by 4.203 C. It would take a limit 4.203 C higher, or a bound 4.203 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.904 C against a 18.0 C limit, so it fails by 4.904 C. It would take a limit 4.904 C higher, or a bound 4.904 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.916 C against a 18.0 C limit, so it fails by 4.916 C. It would take a limit 4.916 C higher, or a bound 4.916 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.662 C against a 18.0 C limit, so it fails by 4.662 C. It would take a limit 4.662 C higher, or a bound 4.662 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.081 C against a 18.0 C limit, so it fails by 4.081 C. It would take a limit 4.081 C higher, or a bound 4.081 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.726 C against a 18.0 C limit, so it fails by 2.726 C. It would take a limit 2.726 C higher, or a bound 2.726 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.811 C against a 18.0 C limit, so it fails by 1.811 C. It would take a limit 1.811 C higher, or a bound 1.811 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.441 C against a 18.0 C limit, so it fails by 0.441 C. It would take a limit 0.441 C higher, or a bound 0.441 C tighter, to change this hour.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.701 C, which is 2.299 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.335 C, and the margin is measured, not chosen: 1.189 C of group-conditional forecast error for this hour of day, plus 0.1463 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.726 C, which is 2.274 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.120 C, and the margin is measured, not chosen: 0.953 C of group-conditional forecast error for this hour of day, plus 0.1666 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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